

IMO Training – Number Theory

Divisibility, Euclidean Algorithm and Modular Arithmetic

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Notes

Notation.

$\mathbb{N}$ refers to the set of natural numbers. $(1, 2, 3, \dots)$

$\mathbb{Z}$ refers to the set of integers. $\mathbb{Q}$ refers to the set of rational numbers. $\mathbb{R}$ refers to the set of real numbers. $\mathbb{C}$ refers to the set of complex numbers.

$\in$ means ‘an element of the set’. For example, $x \in \mathbb{N}$ means ‘ x is a natural number’.

$\forall$ means ‘for all’, and $\exists$ means ‘there exists’. For example, $\forall n \in \mathbb{N}, 2n + 1$ is odd.

Definition. For $a, b \in \mathbb{Z}$, with $b \neq 0$, b divides a if $\exists m \in \mathbb{Z}$ such that $a = mb$. If this holds, we write $b \mid a$. Otherwise, we write $b \nmid a$.

Definition. (Greatest Common Divisor) For nonzero $a, b \in \mathbb{Z}$, $d \in \mathbb{N}$ is the *greatest common divisor* (G.C.D.) of a, b (written as $d = (a, b)$) if

1. $d \mid a$ and $d \mid b$;
2. $\forall c \in \mathbb{Z}$, if $c \mid a$ and $c \mid b$, then $c \mid d$.

Definition. (Modular Arithmetic) For $a, b, r \in \mathbb{Z}$, $b > 0$, $a \equiv r \pmod{b}$ iff (if and only if) $b \mid a - r$. Thus, b divides a is equivalent to $a \equiv 0 \pmod{b}$.

Proposition. (Euclidean Algorithm) For $a, b, r \in \mathbb{Z}$, $b > 0$, with $a > b$, $b \nmid a$ and $a \equiv r \pmod{b}$, we have $(a, b) = (r, b)$.

Definition. (Inverse) For $a, b \in \mathbb{Z}$, $b > 0$, we say a is *invertible* modulo b if $\exists m \in \mathbb{Z}$ such that $am \equiv 1 \pmod{b}$. We call such m the *inverse* of a modulo b . Such inverse, if exists, is also unique up to modulo b .

Theorem. (Chinese Remainder Theorem) For $n \in \mathbb{N}$, with $m_1, m_2, \dots, m_n \in \mathbb{N}$ mutually coprime, then the system of congruence equations

$$\begin{cases} x \equiv x_1 \pmod{m_1} \\ x \equiv x_2 \pmod{m_2} \\ \vdots \\ x \equiv x_n \pmod{m_n} \end{cases}$$

is solvable uniquely up to modulo $m_1 m_2 \dots m_n$.

Example. Find all $n \in \mathbb{Z}$ for which $n^2 + n + 1 \mid n^3 + 2n^2 - n - 3$.

Example. Find all $a, b \in \mathbb{N}$ such that $a \mid b + 1$ and $b \mid a + 1$.

Example. Let $a, b \in \mathbb{N}$ such that $a \mid b^2 + 1$ and $b \mid a^2 + 1$. Prove that $ab \mid a^2 + b^2 + 1$.

Example. Prove that $8 \mid n^2 - 1$ for any odd integer n .

Example. For positive integer $n > 1$, find $(n^{120} - 1, n^{35} - 1)$.

Example. Prove that $n^3 + n^2 + 2n + 1$ and $n^2 + n + 1$ are relatively prime for any $n \in \mathbb{Z}$.

Example. Let $a, b \in \mathbb{N}$ be coprime. Show that there are no $m, n \in \mathbb{N}$ s.t. $am + bn = ab$.

Example. Prove that $\forall n \in \mathbb{Z}, n^3 \equiv n \pmod{6}$.

Example. Solve the following congruence equations:

(a) $3n \equiv 4 \pmod{7}$

(b) $4n \equiv 1 \pmod{6}$

(c) $15n \equiv 5 \pmod{10}$

Example. Solve the following systems of congruence equations:

$$(a) \begin{cases} n \equiv 2 \pmod{3} \\ n \equiv 5 \pmod{7} \end{cases}$$

$$(b) \begin{cases} 3n \equiv 4 \pmod{17} \\ 2n \equiv 3 \pmod{19} \end{cases}$$

$$(c) \begin{cases} 2n \equiv 2 \pmod{3} \\ 3n \equiv 5 \pmod{4} \\ 2n \equiv 3 \pmod{5} \end{cases}$$

Exercises

1. (IMO 1959 Problem 1) Prove that the fraction $\frac{21n+4}{14n+3}$ is irreducible for every $n \in \mathbb{N}$.
2. Let a, b be integers such that $(a, b) = 1$. Prove that
 - (a) $(a+b, a-b) = 1$ or 2
 - (b) $(a^2+b^2, a+b) = 1$ or 2
 - (c) $(a+b, a^2-3ab+b^2) = 1$ or 5
3. Let a, b be integers such that $(a, b) = 1$.
 - (a) Find (a^3+b^3, a^3-b^3)
 - (b) Find (a^2-b^2, a^3-b^3)
4. Let $m, n \in \mathbb{Z}$ s.t. $17 \mid 2m+3n$. Prove that $17 \mid 9m+5n$.
5. Suppose $b \mid a$. Prove that $2^b - 1 \mid 2^a - 1$.
6. Prove that integers $2^n - 1$ and $2^n + 1$ cannot be both primes.
7. (IMO 1975 Problem 4) When 4444^{4444} is written in decimal notation, the sum of its digits is A . Let B be the sum of the digits of A . Find the sum of digits of B .
8. Find all $n \in \mathbb{Z}$ s.t. the last three digits of n^2 are 144 in that order.

Extra Space